

VoIP – Basics and Protocols

CEN 223 - Internet Communication

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What is VoIP?

Transmission of **voice** and **signaling**
via data networks

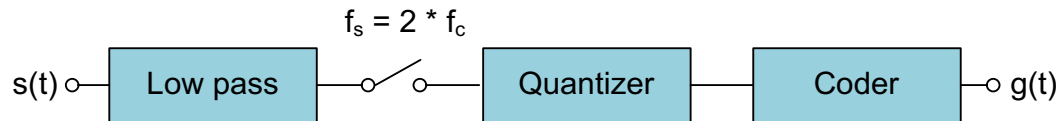
using Internet Protocol (IP),
i.e. via so-called **IP networks** (Internet,
Intranets)

Basics of how it works

- ▶ We digitalize audio with an ADC (analog to digital converter)
- ▶ Transmit it over a digital network (Internet)
- ▶ Transform it again in analog format with DAC (digital to analog converter)
- ▶ In other words, digitalizing voice in data packets, sending them and reconverting them in voice at destination
- ▶ Why digitize?
 - ▶ We can compress it
 - ▶ We can route it
 - ▶ We can convert it to a new better format (more compressed with similar or no quality loss to the human ear as the previous format)

Analog to Digital Conversion (1)

- ▶ This is made by hardware. Today every sound card allows you convert with 16 bit a band of 22050 Hz.
- ▶ The objective is that a VoIP terminal ultimately generates a digital signal from an analog sound signal.
- ▶ As input, the VoIP terminal receives a sound signal $s(t)$, which is of an analog nature, and provides at the output a signal $g(t)$, which is in binary form.



f_s : Sampling frequency
 f_c : Cut-off frequency

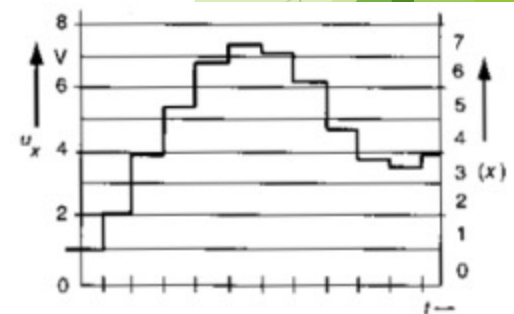
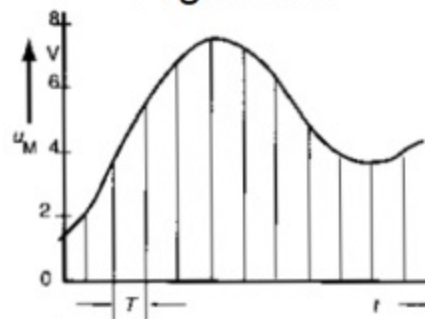
Conversion of an analog signal into a digital signal.

Analog to Digital Conversion (2)

- ▶ If an analog signal $s(t)$ arrives, first a band limitation takes place. Particularly, the analog signal is narrowed in its frequency spectrum, provided that it can be reconstructed as clearly as possible afterward.
- ▶ This band limitation is caused by an element called a low pass. This maximum band-limited signal, also referred to as the cut-off frequency f_c , is determined by speech intelligibility.
- ▶ The conventional telephony has a **cutoff frequency of 4 kHz**.

Analog to Digital Conversion (3)

- ▶ The steps to be taken to convert this band-limited signal $s(t)$ into a binary stream $g(t)$ can be outlined as follows:
 - **Sampling:** The first step is to convert the signal from a time- and value-continuous form into a time-discrete form. For this purpose, the analog signal is sampled at twice the cutoff frequency f_c .
 - **Quantization:** Thereafter, the sampled signal must be quantized to convert it from the time-discrete, but value-continuous form into a time- and value-discrete form.
 - **Coding and compression:** The time- and value-discrete signal must finally be coded and, if necessary, compressed.

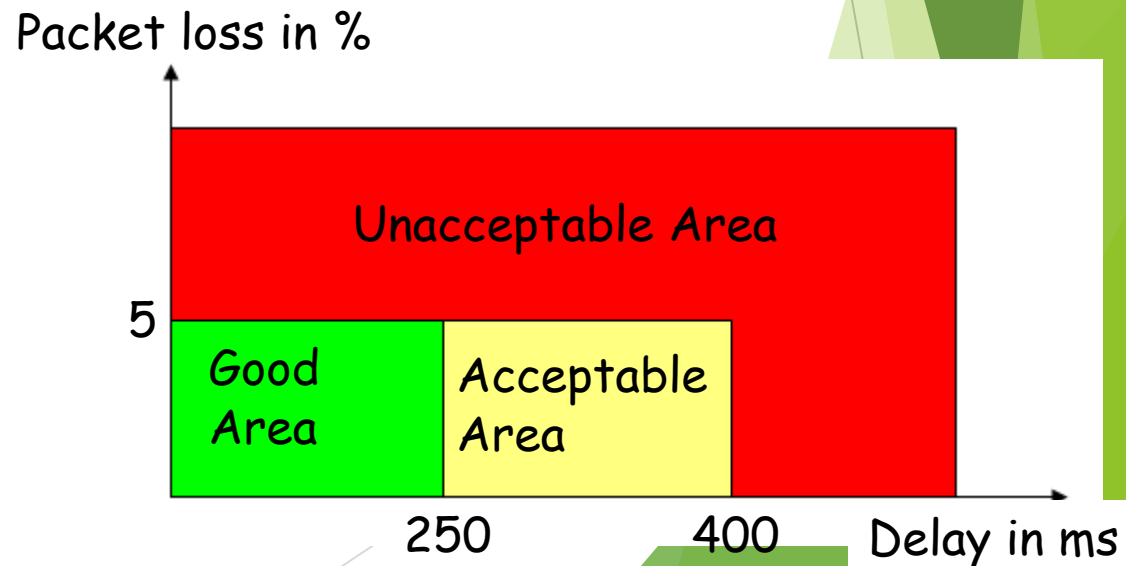


Compression Algorithms

- ▶ Voice bandwidth is 4 kHz, so sampling bandwidth has to be 8 kHz (**according to Nyquist-Shannon sampling theorem**)
- ▶ We represent each sample with 8 bit (having 256 possible values).
- ▶ Throughput is $8000 \text{ Hz} * 8 \text{ bit} = 64 \text{ kbit/s}$, as a typical digital phone line.
- ▶ PCM, Pulse Code Modulation, Standard ITU-T G.711
- ▶ (North America) mu-law and (Europe) a-law variants code analog signal on a logarithmic scale using 12 or 13 bits instead of 8 bits

Requirements for VoIP networks

- ▶ Requirements from the user's point of view
 - ▶ Voice quality
 - ▶ Availability and Reachability
 - ▶ Eavesdropping security
 - ▶ Performance characteristics
 - ▶ Flexibility
- ▶ Requirements from a technical point of view
 - ▶ Delay
 - ▶ Jitter
 - ▶ Echo
 - ▶ Packet losses
 - ▶ Distortions



VoIP Signaling (1)

- ▶ Signaling
 - ▶ Session Initiation Protocol (SIP), [IETF]
 - ▶ H323, [ITU]
 - ▶ Skinny Client Control Protocol (SCCP), [Cisco]
 - ▶ Inter Asterisk Exchange Protocol (IAX™), [Digium]
- ▶ Protocols for the transmission of multimedia / voice data
 - ▶ Real Time Protocol (RTP) and
 - ▶ Real Time Control Protocol (RTCP), [IETF]

VoIP Signaling (2)

► Components

► User Agent

- „Hard“ or „softphone“ acting as participant equipment
- User Agent Client (UAC) and User Agent Server (UAS)

► Proxy

- Routing for signaling messages
- stateless or statefull)

► Redirect Server

► Registrar Server

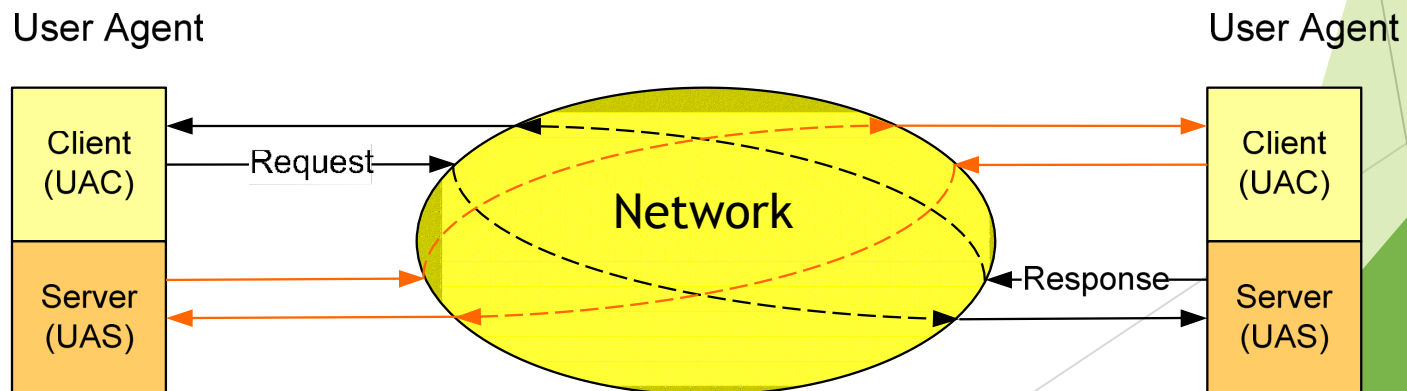
- Participant management
- Participant mobility

► Location Server

- Mapping of permanent to temporary addresses

Session Initiation Protocol (SIP)

- ▶ SIP: Protocol of the application layer, RFC 3261 - 269 pages
- ▶ SIP supports the establishment and operation of so-called sessions for multimedia communication (**voice, video, text**) between Internet users
- ▶ Two or more user agents are involved in a session
- ▶ SIP supports the use of a network infrastructure for registering and localizing participants, receiving connection requests, etc. => Proxy server
- ▶ **SIP uses naming similar to e-mail** (SIP:badri@ceng.cu.edu.tr, SIP:17324452082@cu.edu.tr etc.



SIP uses a request / response model (like HTTP)

Session Initiation Protocol (SIP) (2)

SIP Facets of Establishing and Terminating
Multimedia Communications

User Location

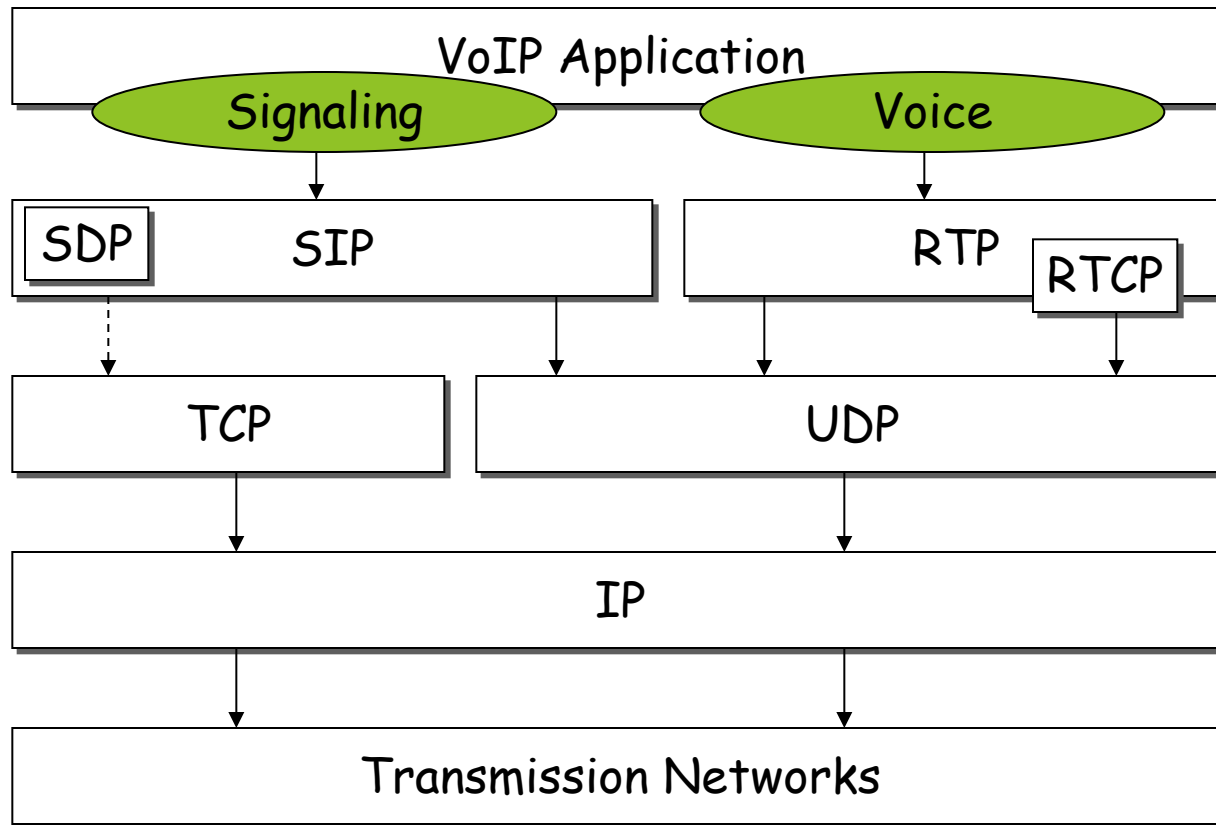
User
Availability

User
Capabilities

Session
Setup

Session
Management

Session Initiation Protocol (SIP) (3)



Classification of SIP and RTP

SIP: Session Initiation Protocol
SDP: Session Description Protocol
RTP: Real Time Transport Protocol
RTCP: Real Time Control Protocol

SIP Request Messages

Request method	Purpose
INVITE	Initiate a call.
ACK	Confirm the final response to an INVITE.
BYE	Terminate a call.
CANCEL	Cancel searches and “ringing”
OPTIONS	Communicate features supported
REGISTERED	Register a client with a location service.

SIP Response Messages

Status-Code	Category	Example information
1xx	Informational	trying, ringing, call is being forwarded, queued
2xx	Success	OK
3xx	Redirection	Moved permanently, moved temporarily, etc
4xx	Client error	Bad request, unauthorized, not found, busy, etc
5xx	Server error	Server error, not implemented, bad gateway, etc
6xx	Global failure	Busy everywhere, does not exist anywhere, etc.

100 Trying

180 Ringing

181 Call is being Forwarded

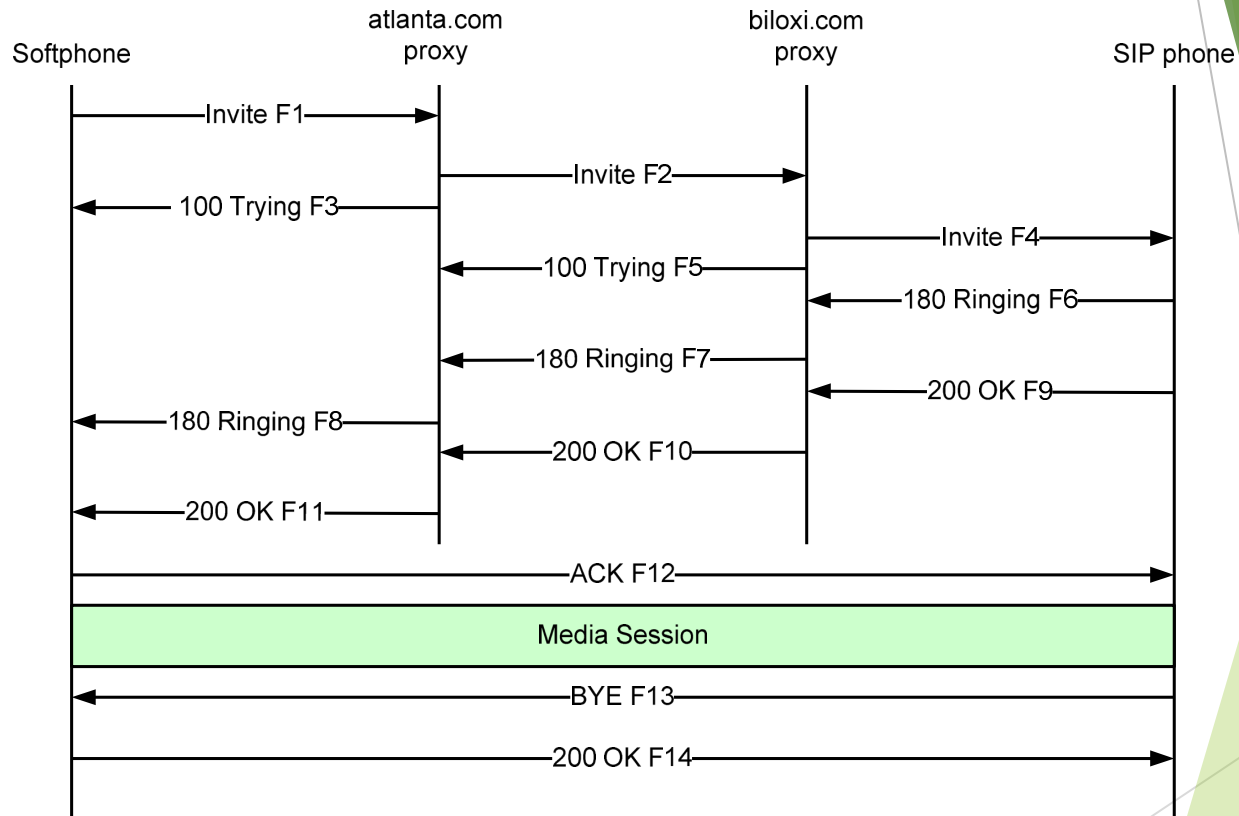
182 Queued

200 OK

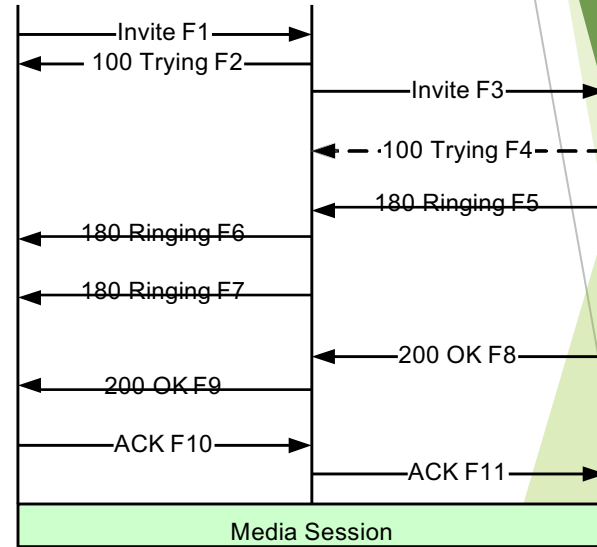
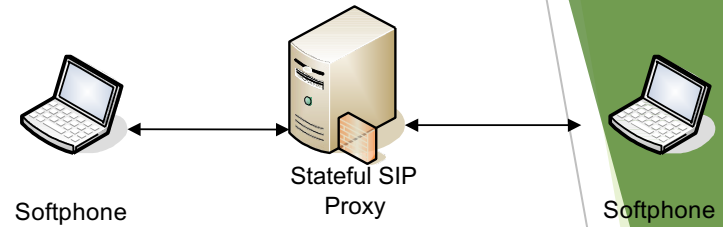
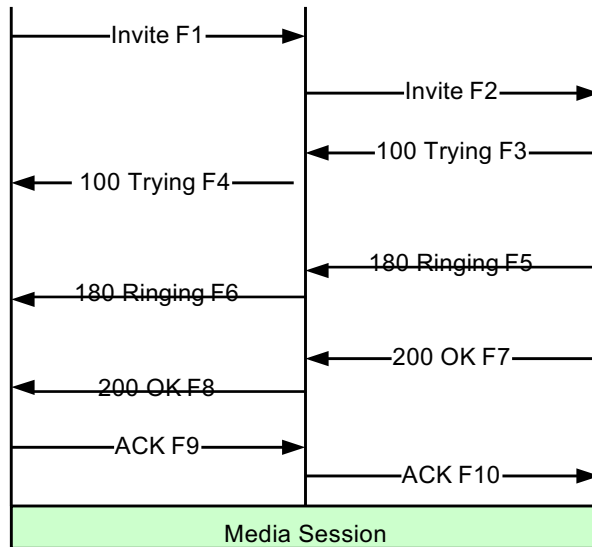
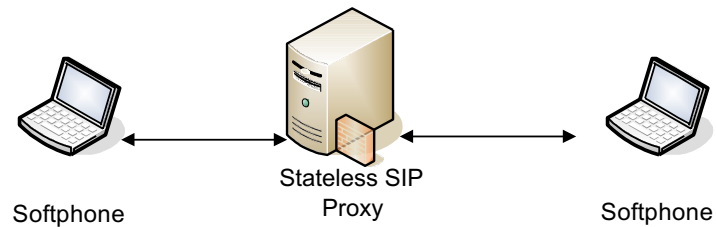
301 Moved Permanently

302 Moved Temporarily

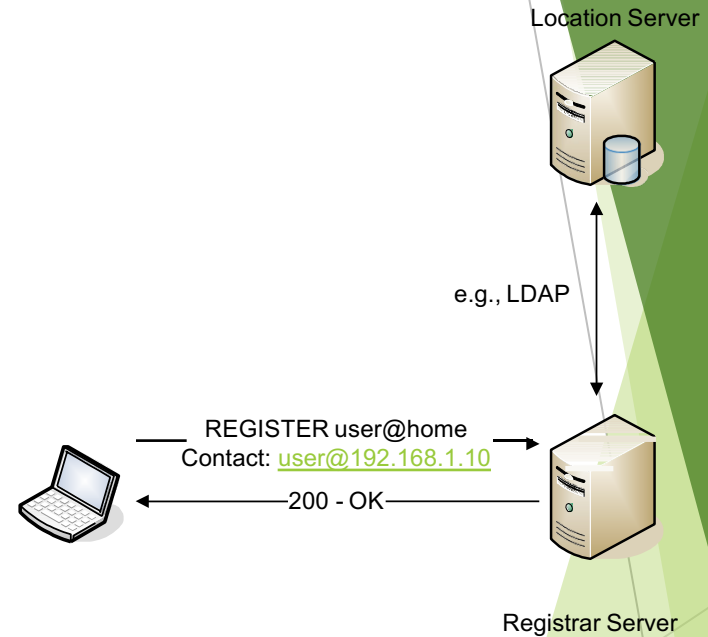
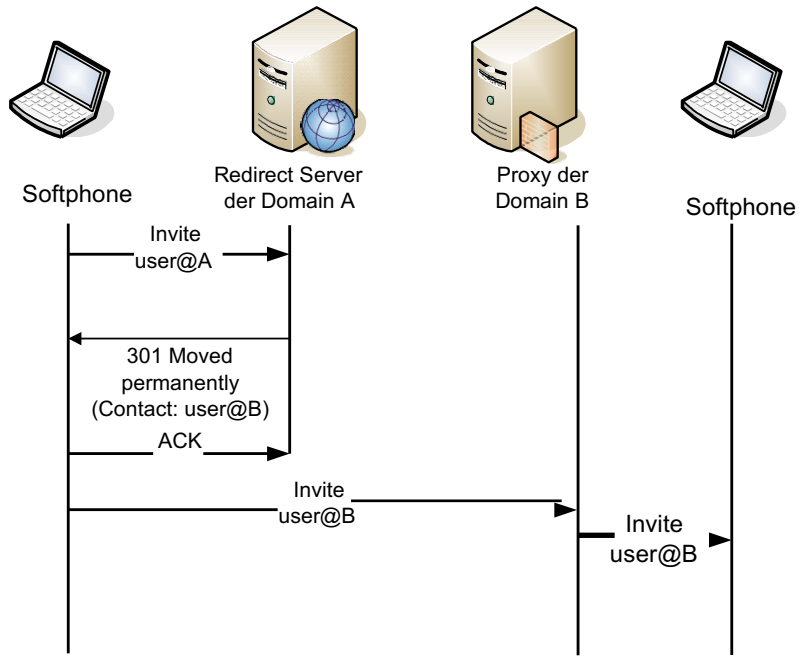
Example: Setting up and triggering a "session"



SIP Network Elements (1)



SIP Network Elements (2)



SIP: Registration & Discovery

Register Request:

Request URI:

domain of the location service
(e.g. sip@chicago.com)

To:

contains address-of-record
(SIP URI)

From:

address-of-record of the person
responsible for the registration

Call-ID:

UAC should use only one CALL-
ID for all registrations to a
particular registrar

Cseq:

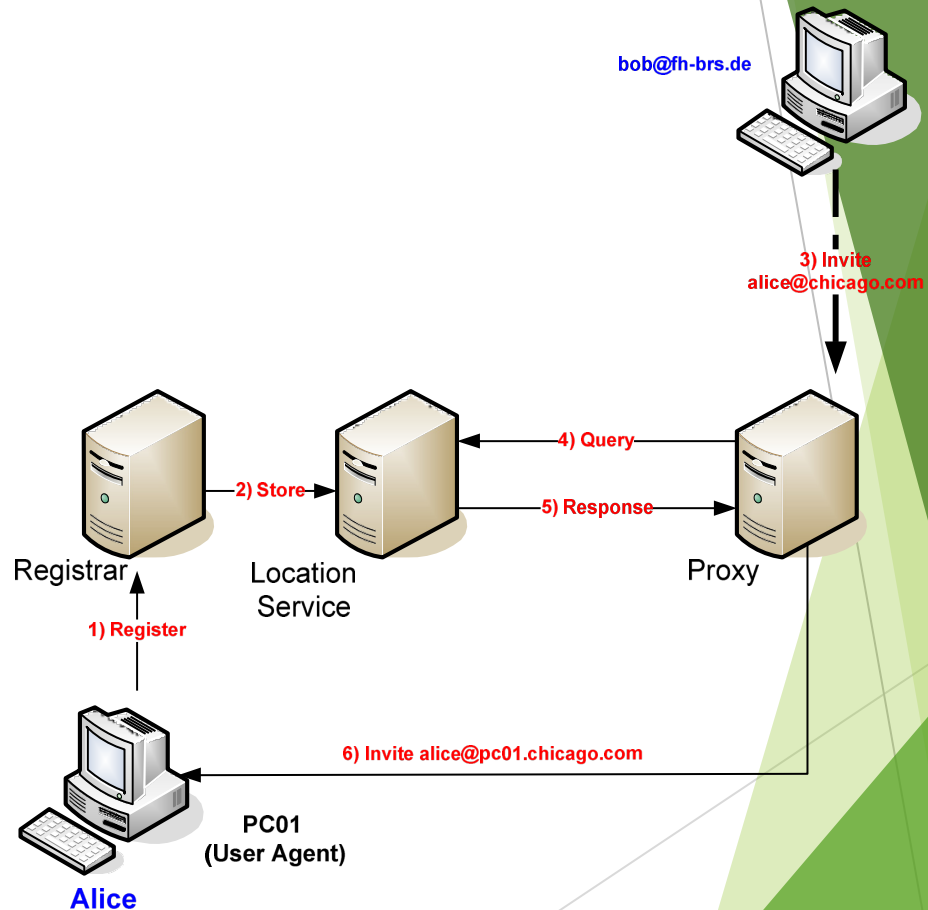
>> sequence number <<

Contact (optional):

address bindings

Expires (optional):

expiration for the binding



SIP - Message

Session Initiation Protocol

Request-Line: INVITE sip:10000@sipgate.de SIP/2.0

Method: INVITE

[Resent Packet: False]

Message Header

Via: SIP/2.0/UDP 192.168.178.102:5060;rport;branch=z9hG4bK60846

Max-Forwards: 70

To: <sip:10000@sipgate.de>

From: <sip:user@sipgate.de>;tag=z9hG4bK66687806

Call-ID: 841479524486@192.168.178.102

CSeq: 1 INVITE

Contact: <sip:user@192.168.178.102>

Expires: 3600

User-Agent: mjsip stack 1.6

Content-Length: 159

Content-Type: application/sdp

Message body

Session Description Protocol

Session Description Protocol Version (v): 0

Owner/Creator, Session Id (o): <sip:user@sipgate.de> 0 0 IN IP4 192.168.178.102

Session Name (s): Session SIP/SDP

Connection Information (c): IN IP4 192.168.178.102

Time Description, active time (t): 0 0

Media Description, name and address (m): audio 21000 RTP/AVP 0

Media Attribute (a): rtpmap:0 PCMU/8000

- ▶ Free Software Clients
 - ▶ Ekiga (Unix, Windows)
 - ▶ Sipgate (Windows)
 - ▶ Kphone (Unix)

Real-Time Transport Protocol (RTP)

- ▶ Why not TCP?
 - ▶ TCP offers reliable, in-sequence data transfer
 - ▶ TCP embeds flow/error/congestion control
- ▶ Why not UDP?
 - ▶ Missing sequence control and not reliable!
- ▶ **RTP:** transport protocol for multimedia traffic. It carries real-time data.
 - ▶ Application level framing; integrated layer processing
 - ▶ Usually RTP/UDP/IP, or RTP/IP
 - ▶ RTP does NOT guarantee real-time itself!
- ▶ **RTP Control Protocol (RTCP):** It monitors the quality of service and conveys information about the participants.

RTP Applications (1)

► Simple Multicast Audio Conference

- The audio-conferencing application used by each conference participant sends audio data **in small chunks of, say, 20 ms duration**.
- Each chunk of audio data is preceded by **an RTP header**; RTP header and data are in turn contained **in a UDP packet**.
- The RTP header indicates what **type of audio encoding** (such as PCM, ADPCM or LPC) is contained in each packet.
- RTP header contains **timing information** and a **sequence number** that allow the receivers to reconstruct the timing produced by the source.
- The sequence number can also be used by the receiver to estimate how many packets are being lost.
- The audio application in the conference periodically multicasts a **reception report** plus the name of its user on the RTCP port. The reception report indicates how well the current speaker is being received.
- A site sends the RTCP **BYE packet** when it leaves the conference.

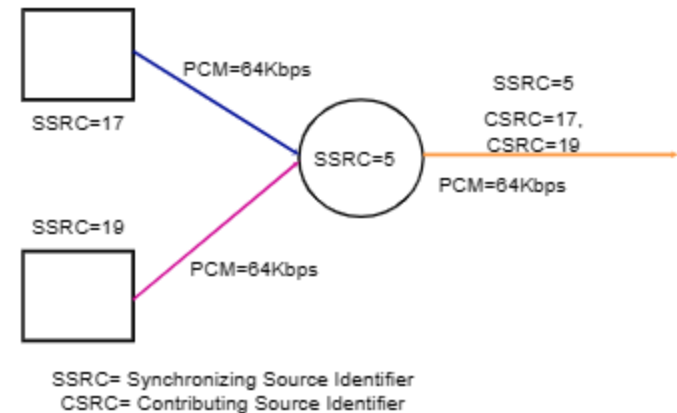
RTP Applications (2)

► Audio and Video Conference

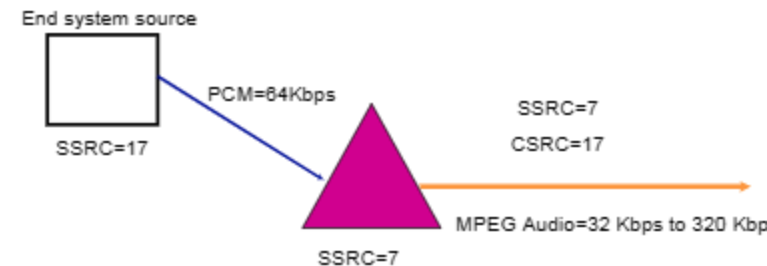
- Audio and video media are transmitted **as separate RTP session** and RTCP packets are transmitted for each medium using two different UDP port pairs and/or multicast addresses.
- There is no direct coupling at the RTP level between the audio and video sessions, except that a user participating in both sessions should use the same distinguished (canonical) name in the RTCP packets for both so that the sessions can be associated.
- Despite the separation, **synchronized playback of a source's audio and video can be achieved using timing information carried in the RTP packets for both sessions.**

RTP Translators and Mixers

- ▶ A mixer combines several media stream into a one new stream (with possible new encoding)
 - ▶ reduced bandwidth networks (video or telephone conference)
 - ▶ appears as new source, with own identifier

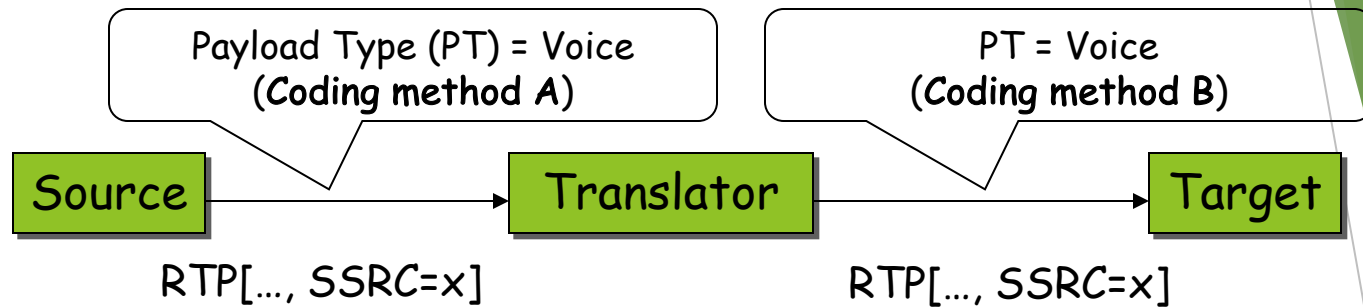


- ▶ A translator changes encoding from one format to another single media stream
 - ▶ may convert encoding



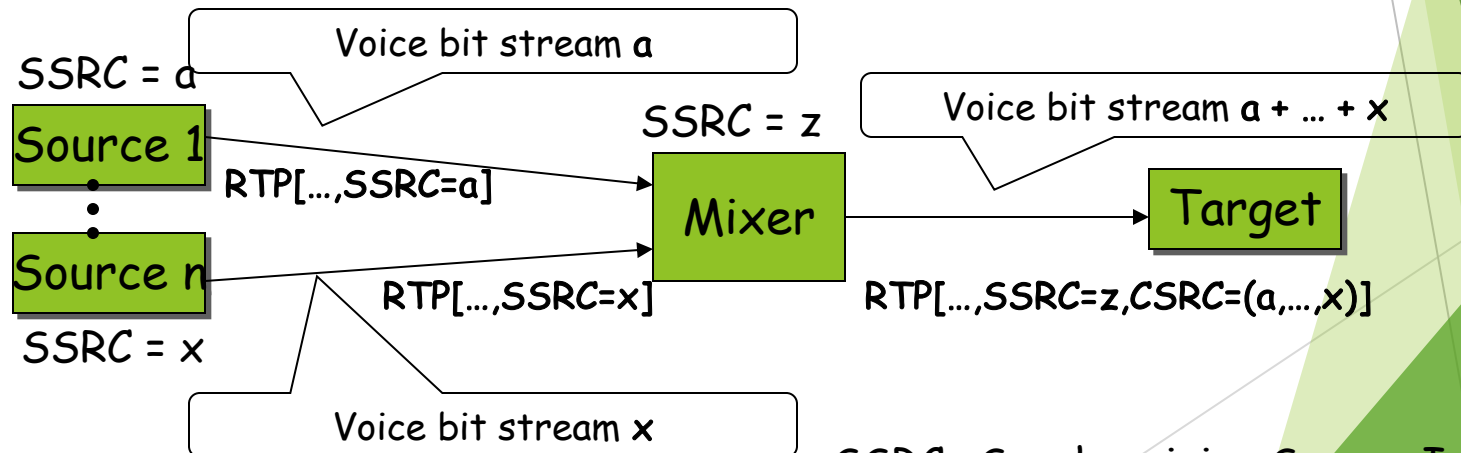
Translator und Mixer

Translator



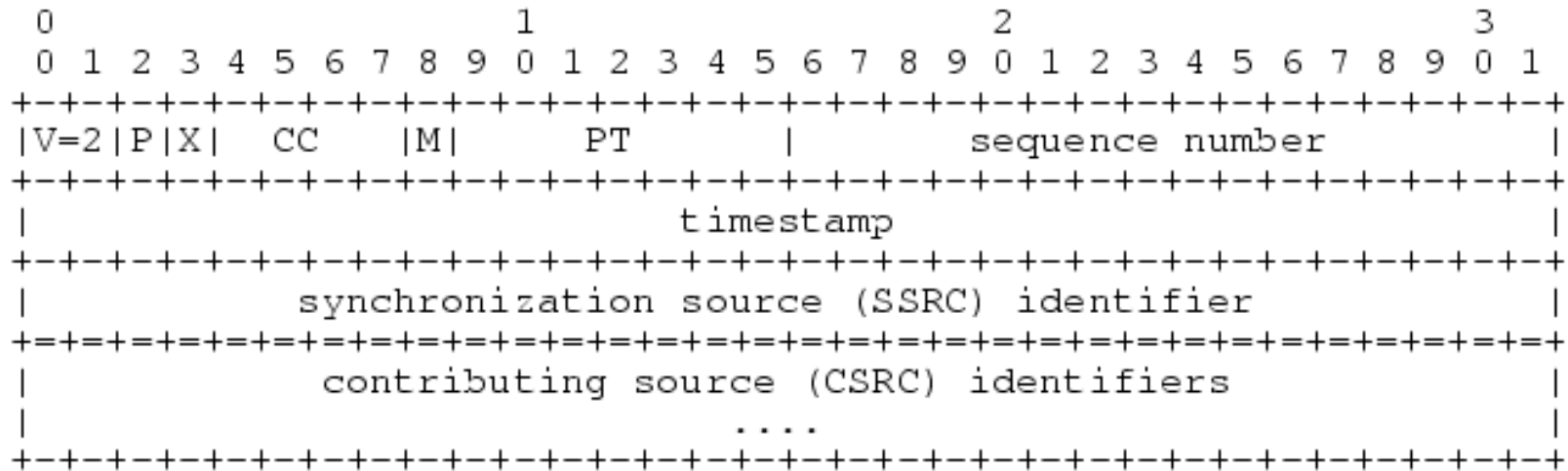
A - Law
 μ - Law

Mixer



SSRC= Synchronizing Source Identifier
CSRC= Contribution Source Identifier

Real-Time Transport Protocol



V: Version

P: Padding

X: Extension

CC: Contribution Source Identifier Count

M: Marker

RTP Data Transfer Protocol (2): Synchronization

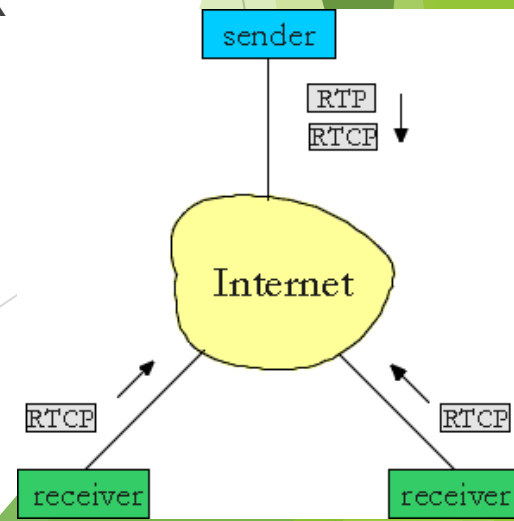
- ▶ Synchronization Source (SSRC)
- ▶ Sequence number (packet count)
 - ▶ gap in sequence #: packet loss
- ▶ Timestamp (sample count)
 - ▶ e.g., audio@8KHz, 20ms samples/packet
 - ▶ timestamp increment per packet: 160
 - ▶ 90KHz used for video
 - ▶ a video frame may be encapsulated in a few packets
 - ▶ gap in timestamp: silence

RTP Data Transfer Protocol (3): RTP profiles

- ▶ Media specific (e.g., audio)
 - ▶ Marker: e.g., the start of a talk spurt
 - ▶ Payload Type: e.g., specific audio codec
 - ▶ PT=0: uPCM 64Kbps; PT=3: GSM 13Kbps
 - ▶ timestamp: e.g., sampling rate, 8KHz PCM
 - ▶ packet size: e.g., about 20ms samples in PCM
 - ▶ packets independent as much as possible: ALF
 - ▶ other issues: e.g., mixed audio channels

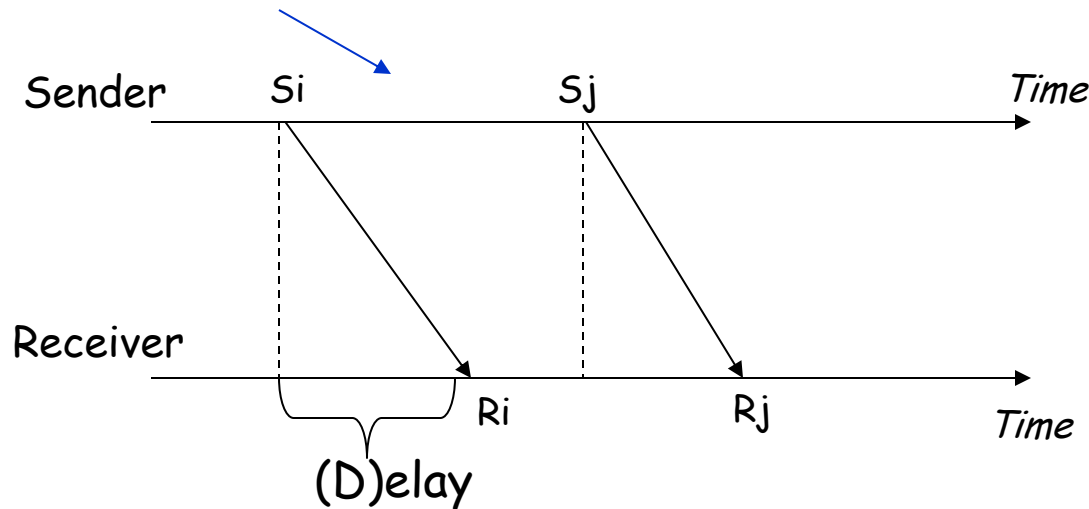
Real-Time Control Protocol (RTCP)

- ▶ RTCP: RTP's control companion
 - ▶ Purpose: feedback control information
 - ▶ for flow/error/congestion/quality control
 - ▶ Two consecutive UDP ports for RTP and RTCP
 - ▶ Sender report: offer sending/reception statistics
 - ▶ Receiver report: offer reception statistics
- ▶ RTCP common header: Version (2-bit); Padding (1); RR count (5); Type (8); Length (16)



Real-Time Control Protocol (RTCP)

- ▶ An integral part of RTP
- ▶ Current properties of the RTP packet stream, e.g. jitter, packet loss
→ QoS monitoring

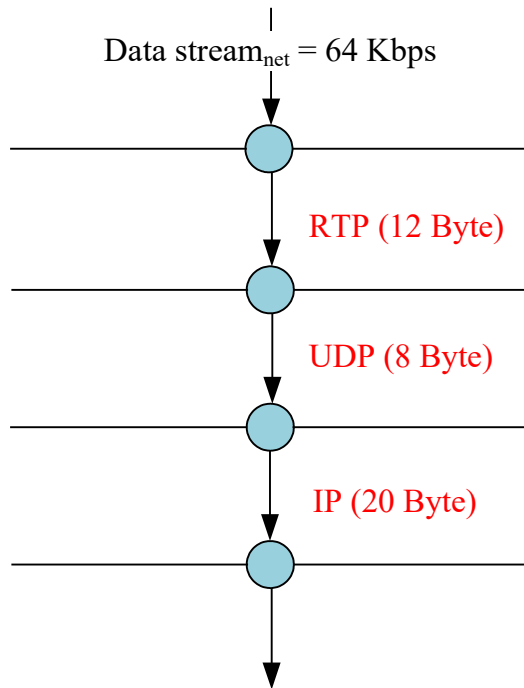


- **Inter-arrival time ($D(i,j)$)** is the time difference between the arrival times of two consecutive packets
- **Jitter ($J(i)$)** is defined as a statistical variance of the RTP data packet inter-arrival time

$$D(i,j) = (R_j - R_i) - (S_j - S_i) = (R_j - S_j) - (R_i - S_i)$$

$$J(i) = J(i-1) + (|D(i-1,i)| - J(i-1))/16$$

Typical VoIP Traffic Overhead



$$f_c = 4 \text{ kHz}$$

$$f_s = 8 \text{ kHz}$$

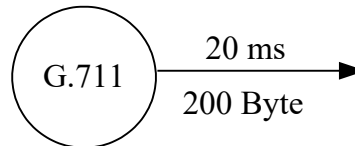
Coding: A-law à 8 bit per sample value

$$\Rightarrow \text{Data stream}_{\text{net}} = 64 \text{ kbps}$$

$$\Rightarrow \text{Packetization time: } 20 \text{ ms}$$

$$\Rightarrow \text{Packet size} = 160 \text{ Byte (without header)}$$

$$\Rightarrow \text{Packet size} = 160 \text{ Byte} + 40 \text{ Byte} = 200 \text{ Byte (including header)}$$



Additional overhead caused by TCP/IP protocol stack regarding VoIP

=> The overhead of 40 Byte in relation to 160 Byte, which one would only have to transfer in the classical telephony, shows the immense overhead caused by VoIP.

Advantages of VoIP

- ▶ Reduced cost
 - ▶ Don't have to pay for line usage
- ▶ Mobility
 - ▶ Can use same identity (same number or user name) from anywhere in the world
- ▶ Other data exchange and synchronization
 - ▶ Can exchange data such as images, graphs, videos and contact info while talking

Disadvantages of VoIP

- ▶ Privacy and security etc is complicated
- ▶ Internet services don't work during power outages
 - ▶ Regular phone uses low DC current and an AC current during ring. 99% uptime even during power outage and/or natural disaster.
- ▶ Some devices such as traditional faxes most likely won't work
- ▶ Emergency calls?